

Greenfire Resources

Deep-tier diligence memo · 2026-06-24

36 FINDINGS	1 SOURCES	2022– 2023 PERIOD	Elevated TOP RISK BAND	Low CONFIDENCE
----------------	--------------	-------------------------	---------------------------	-------------------

BOTTOM LINE

Executive summary

Searched interpreted due-diligence findings on Greenfire Resources, focused on regulatory emissions/release records for the Greenfire Resources Operating Corporation Hangingstone SAGD Demonstration Facility, 2022 reporting period. Records reviewed cover substance-specific release disclosures: hydrogen sulphide (176362), total reduced sulphur (176363), ethylene glycol (176364), methanol (176365), toluene (176366), benzene (176367), xylene/all isomers (176368), and total VOCs (176369), consistent with a Canadian multi-pollutant reporting regime (likely NPRI). Coverage is thin and limited in several...

WHAT WE FOUND	WHY IT MATTERS
Multi-pollutant emissions reporting profile at Hangingstone SAGD Demonstration Facility (2022)	Greenfire Resources Operating Corporation filed a cluster of substance-specific release records for its Hangingstone SAGD Demonstration Facility covering the 2022 reporting period. The substances span sour-gas/sulphur compounds (hydrogen sulphide, total reduced sulphur), process chemicals (ethylene glycol, methanol), and BTEX/VOC compounds (benzene, toluene, xylene, ethylbenzene, n-hexane,... source_records:176362 source_records:176363 source_records:176364 +5
Reported release of regulated toxic and sour-gas substances	The facility's 2022 disclosure includes substances that are regulated as toxic or hazardous under Canadian environmental frameworks, notably benzene (a known carcinogen) and hydrogen sulphide / total reduced sulphur (sour gas hazards). Their presence in the dataset indicates the facility was subject to reporting obligations... source_records:176362 source_records:176363 source_records:176367

RISK SNAPSHOT

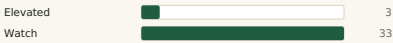
Where the exposure concentrates

Political & reputational

Elevated · 36 finding(s)

Not observed in this run: legislative & regulatory, government support & dependencies, lobbying & stakeholders.

SEVERITY MIX



ACTIVITY OVER TIME



SECTOR EXPOSURE



WHAT'S MOVING

Material developments

Greenfire Resources Operating Corporation — Carbon monoxide (2023) — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Carbon monoxide (dated 2023-01-01) 2023-01-01

Without the actual emission quantity, it is not possible to benchmark the facility's CO releases against regulatory thresholds, industry peers, or year-over-year trends. The breadth of pollutants reported across the contextual records indicates the facility has multiple regulated air release streams, which may be material to environmental liability assessments, regulatory compliance reviews, and ESG disclosure obligations relevant to a transaction or investment. [source_records:295578](#)

Greenfire Resources Operating Corporation — Carbon monoxide (2022) — Hangingstone SAGD Demonstration Facility: released [amount not specified] tonnes of Carbon monoxide (dated 2022-01-01) 2022-01-01

Without a confirmed emissions quantity, it is not possible to assess the scale of CO releases relative to regulatory thresholds or peer benchmarks. The absence of a quantity value limits any assessment of potential compliance exposure, liability, or reputational risk associated with this specific pollutant. The broader pattern of multi-substance reporting across the same facility and year suggests ongoing regulatory disclosure obligations that a counterparty should be aware of. [source_records:176378](#)

Greenfire Resources Operating Corporation — Carbon monoxide (2022) — Hangingstone Expansion Project: released [amount not specified] tonnes of Carbon monoxide (dated 2022-01-01). 2022-01-01

Incomplete quantity data limits the ability to characterize the CO release's scale, regulatory standing, or trend. If the actual tonnage is material, it could affect the company's environmental compliance profile, permitting conditions, or future carbon/air-quality regulatory obligations — each of which may be relevant to operational risk assessment. The breadth of substances reported across both Hangingstone facilities may indicate a complex air-emissions management obligation warranting closer review. [source_records:185038](#)

Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [quantity not populated] tonnes of Hydrochloric acid, dated 2023-01-01. 2023-01-01

The inability to assess release quantity prevents any calibration of regulatory exposure, potential enforcement risk, or compliance standing. A material Hydrochloric acid release could attract regulatory scrutiny, community opposition, or remediation obligations depending on volume and site conditions. The breadth of co-reported substances across the same facility increases the surface area of potential environmental liability that a counterparty may wish to investigate. [source_records:295570](#)

Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Nitrogen oxides (expressed as nitrogen dioxide) in 2023, as recorded in a source record dated 2023-01-01. The quantity field (amount) is null/unpopulated. 2023-01-01

Without the quantified tonnage, it is not possible to assess the materiality of the facility's NOx emissions relative to applicable regulatory thresholds (e.g., Canada's National Pollutant Release Inventory reporting thresholds), potential compliance obligations, or environmental liability exposure. The breadth of missing quantities across all co-reported substances amplifies this gap and limits the ability to characterize the facility's overall environmental footprint from this source alone. [source_records:295571](#)

Greenfire Resources Operating Corporation — Methanol (2023) — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Methanol (dated 2023-01-01). 2023-01-01

Without the actual release quantities, it is not possible to assess the magnitude of environmental liabilities, compliance standing, or potential regulatory exposure associated with methanol releases at this facility. The breadth of substances reported across the same facility and period indicates that environmental disclosure obligations are active and multi-dimensional, which may be material to environmental, social, and governance (ESG) assessments, permit conditions, or stakeholder engagement considerations. [source_records:295572](#)

+30 further finding(s) in the evidence appendix.

SO WHAT

Material risks & opportunities

MATERIAL RISKS

OPPORTUNITIES & ANGLES

- The Hangingstone SAGD Demonstration Facility reported releases of regulated toxic substances including benzene and sour-gas compounds (hydrogen sulphide, total reduced sulphur) in 2022, indicating exposure to environmental and health-related regulatory reporting obligations.
[source_records:176362](#) [source_records:176363](#)
[source_records:176367](#)
- The magnitude of every reported release cannot be assessed because the quantity/tonnage field is null or blank across all retrieved substance records. This prevents any evaluation of whether releases were de minimis, below reportable thresholds, or material, and limits the... [source_records:176362](#) [source_records:176363](#)
[source_records:176364](#) +5
- The blank quantity fields may reflect either data-omission/population gaps in the source dataset or releases falling below reportable thresholds; the cause is undetermined, creating uncertainty about the completeness and reliability of the emissions disclosure as captured. [source_records:176365](#)
[source_records:176369](#)

None identified in this run.

WHO

Political stakeholders & connections

[Standing Committee on Environment and Sustainable Development](#)

COMMITTEES

[Environment](#)

ORGANIZATIONS

NEXT

Diligence actions

QUESTIONS FOR MANAGEMENT

1. What are the actual reported tonnages for each substance (H₂S, total reduced sulphur, benzene, toluene, xylene, ethylbenzene, n-hexane, methanol, ethylene glycol, total VOCs) at the Hangingstone SAGD Demonstration Facility for 2022, and how do they compare to reportable thresholds?
2. Are the blank quantity fields due to sub-threshold releases, dataset population gaps, or non-reporting? Can the underlying NPRI/provincial filings be obtained to confirm?
3. How do the 2022 emissions compare to prior and subsequent reporting years to establish a trend for the facility?
4. Are there any compliance actions, exceedances, or enforcement records associated with these reported substances at the facility?

5. Do the additional retrieved records (185032-185038, 295570-295580) relate to the same facility, other Greenfire assets, or different topics, and what do they cover?
6. Which specific regulatory regime governs these disclosures (federal NPRI vs. Alberta provincial reporting), and what are the associated permit/license conditions?

EVIDENCE

Appendix & coverage

SOURCE COVERAGE

Breadth sources

36

SOURCE	RISK	RECORD	DATE	
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [blank/null] tonnes of Hydrogen sulphide, dated 2022-01-01.	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Total reduced sulphur (expressed as hydrogen sulphide) (2022) — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Total reduced sulphur (expressed as hydrogen sulphide), dated 2022-01-01.	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Ethylene glycol (2022) — Hangingstone SAGD Demonstration Facility: released tonnes of Ethylene glycol (dated 2022-01-01). The release quantity field is null/blank in the record.	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Methanol, dated 2022-01-01.	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Toluene, dated 2022-01-01.	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Benzene (2022) — Hangingstone SAGD Demonstration Facility: released tonnes of Benzene (dated 2022-01-01)	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [quantity not specified] tonnes of Xylene (all isomers), dated 2022-01-01.	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Volatile Organic Compounds (Total) (2022) — Hangingstone SAGD Demonstration Facility: released [amount not populated] tonnes of Volatile Organic Compounds (Total) (dated 2022-01-01).	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [blank/null] tonnes of PM2.5 - Particulate Matter <= 2.5 Micrometers, dated 2022-01-01.	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — PM10 - Particulate Matter <= 10 Micrometers (2022) — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of PM10 - Particulate Matter <= 10 Micrometers (dated 2022-01-01).	2022-01-01	open
	WATCH		2022-01-01	open

SOURCE	RISK	RECORD	DATE	
Breadth sources		Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [amount not populated] tonnes of Total particulate matter, dated 2022-01-01.		
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [quantity not recorded] tonnes of Ethylbenzene, dated 2022-01-01.	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — n-Hexane (2022) — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of n-Hexane (dated 2022-01-01).	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of 1,2,4-Trimethylbenzene, dated 2022-01-01.	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Sulphur dioxide, dated 2022-01-01.	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [quantity not specified] tonnes of Nitrogen oxides (expressed as nitrogen dioxide) in 2022, as recorded on 2022-01-01.	2022-01-01	open
Breadth sources	ELEVATED	Greenfire Resources Operating Corporation — Carbon monoxide (2022) — Hangingstone SAGD Demonstration Facility: released [amount not specified] tonnes of Carbon monoxide (dated 2022-01-01)	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hydrochloric acid (2022) — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Hydrochloric acid (dated 2022-01-01).	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Methanol (2022) — Hangingstone Expansion Project: released [blank] tonnes of Methanol (dated 2022-01-01)	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Volatile Organic Compounds (Total) (2022) — Hangingstone Expansion Project: released [amount not specified] tonnes of Volatile Organic Compounds (Total) (dated 2022-01-01).	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — PM10 - Particulate Matter <= 10 Micrometers (2022) — Hangingstone Expansion Project: released [blank] tonnes of PM10 - Particulate Matter <= 10 Micrometers (dated 2022-01-01)	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — PM2.5 - Particulate Matter <= 2.5 Micrometers (2022) — Hangingstone Expansion Project: released [blank] tonnes of PM2.5 - Particulate Matter <= 2.5 Micrometers (dated 2022-01-01)	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone Expansion Project: released [blank] tonnes of Total particulate matter in 2022 (dated 2022-01-01).	2022-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hydrogen sulphide (2022) — Hangingstone Expansion Project: released [blank] tonnes of Hydrogen sulphide (dated 2022-01-01)	2022-01-01	open
	ELEVATED		2022-01-01	open

SOURCE	RISK	RECORD	DATE	
Breadth sources		Greenfire Resources Operating Corporation — Carbon monoxide (2022) — Hangingstone Expansion Project: released [amount not specified] tonnes of Carbon monoxide (dated 2022-01-01).		
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [quantity not populated] tonnes of Hydrochloric acid, dated 2023-01-01.	2023-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Nitrogen oxides (expressed as nitrogen dioxide) in 2023, as recorded in a source record dated 2023-01-01. The quantity field (amount) is null/unpopulated.	2023-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Methanol (2023) — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Methanol (dated 2023-01-01).	2023-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Benzene (2023) — Hangingstone SAGD Demonstration Facility: released [quantity not specified] tonnes of Benzene (dated 2023-01-01)	2023-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — n-Hexane (2023) — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of n-Hexane (dated 2023-01-01)	2023-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of PM10 - Particulate Matter <= 10 Micrometers, dated 2023-01-01.	2023-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Hydrogen sulphide, dated 2023-01-01.	2023-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Volatile Organic Compounds (Total) (2023) — Hangingstone SAGD Demonstration Facility: released [amount not specified] tonnes of Volatile Organic Compounds (Total) (dated 2023-01-01).	2023-01-01	open
Breadth sources	ELEVATED	Greenfire Resources Operating Corporation — Carbon monoxide (2023) — Hangingstone SAGD Demonstration Facility: released [blank] tonnes of Carbon monoxide (dated 2023-01-01)	2023-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Sulphur dioxide (2023) — Hangingstone SAGD Demonstration Facility: released tonnes of Sulphur dioxide (dated 2023-01-01). The quantity field (amount) is null in the record.	2023-01-01	open
Breadth sources	WATCH	Greenfire Resources Operating Corporation — Toluene (2023) — Hangingstone SAGD Demonstration Facility: released tonnes of Toluene (dated 2023-01-01). The quantity field (amount) is null in the record.	2023-01-01	open

Method: AI-assisted (Claude) · 5/6 rounds · 41 model calls · provider claude. Every cited record is independently retrievable at /records/<table>/<pk>.